

UNIT THREE
INTRODUCTION TO LINEAR PROGRAMMING
Definition and Basic Concepts of Linear Programming (LP)

Definition

Linear Programming (LP) is a mathematical optimization technique, which helps managers in making the best use of economic resources. Firm's economic resources: materials, capital, labor, time, space, etc are scarce, i.e., limited. To achieve objective(s) of organization, these resources must be allocated efficiently and combined in different ways, since some combinations are better than others in meeting objectives. Therefore, LP is a powerful and widely applied technique to allocate scarce resources and how to do such an allocation in the best possible way among competing uses subjected to more than one limiting conditions expressed in the form of equations and/ or inequalities.

Note: By "optimization" technique, we mean a method, which attempts to maximize or minimize some objectives, like maximize profit, sales or minimize costs. LP is a subset of larger area of mathematical optimization procedures called mathematical programming.

Basic Concepts

- *Decision Variables* are the set of elements whose values are unknown. These variables are the level of output to be produced or services to be rendered from each type to optimize the objective, since all combinations do not lead to the same optimal result. These variables which need decisions as to their values at available resource is represented by X_i , where i represents variables involved like $(X_1, X_2 \dots X_n)$.
- *Objective function* is a mathematical representation of the overall goal stated as a function of the decision variables, X_i . The basic structure of a LP problem is either to maximize or minimize an objective function while satisfying the set of constraining conditions. The objective function may represent goals such as profit level, total revenue, total cost, market shares, percent return on investment.

Note: Objective function is always single; no two objectives can be attained at the same time at the available resources. It is quantifiable and is designed by $Z_{\max}/Z_{\min}$ or $v_{\max}/v_{\min}$.

- *Operational Constraints* represent a set of constraining conditions which must be satisfied when determining levels for decision variables. They are limiting factors imposed on the objective function. In other words, constraints indicate the total of each type of resources available while producing or rendering the required level of output. Constraints could be either internal (labor hours, machine hours, raw materials, storage space, capital, etc) or external (demand level of a product, work standards, government regulations). Constraints also can be categorized as:
 - i) *System constraint* is a constraint consisting of two or more decision variables.
 - ii) *Individual or unit constraint* is a constraint consisting of only one variable.
 - iii) *Non- negativity constraint* is a constraint which states the value of a variable do not entertain negative numbers

The constraints of LP problem can be represented by equations or by inequalities ($\leq$, and/ or $\geq$ types). These problems are called LP problems because the objective function and constraints are all linear.

Example 3.1

Let X_1 = quantity of product 1
 X_2 = quantity of product 2
 X_3 = quantity of product 3

} decision variables, X_i

Maximize: $4X_1 + 7X_2 + 5X_3$ – Objective function (maximization problem)

$$\text{Structural constraints } \left\{ \begin{array}{l} \text{Subject to:} \\ 2X_1 + 3X_2 + 6X_3 \leq 300 \text{ hours} \\ 2X_1 + 3X_2 \leq 200 \text{ kg} \\ X_3 \leq 30 \text{ units} \end{array} \right\} \text{ system constraints}$$

$X_1, X_2, X_3 \geq 0$ ← non-negativity constraint

- *Parameter* involves coefficients of objective function and operational constraints. For example 4, 7, and 5 of coefficient of objective function, 2, 3, and 6 of the first constraint are parameters.

There are four elements in operational constraints:

i) A Right Hand Side (RHS), which is denoted by RHS is the quantity that specifies the limit for that constraint. It must always be a constant; and not a variable. If a variable exists, it requires rearrangement.

Example 3.2

$$2X_1 + 3X_2 - 300 \leq -6X_3 \quad \leftarrow \text{is a variable and would be rearranged as:}$$

$$2X_1 + 3X_2 + 6X_3 \leq 300$$

ii) An algebraic sign indicating whether RHS quantity is upper or lower bound, which can not be exceeded.

- ✓ If the algebraic sign is $\leq$, the RHS represents the upper bound (i.e., the maximum resource available).
- ✓ If the algebraic sign is $\geq$, the RHS represents the lower bound (the minimum amount available).
- ✓ If the algebraic sign is $=$, the RHS represents the exact available amount of resource.

In the above example, $2X_1 + 3X_2 + 6X_3 \leq 300$ hours, the $\leq$ sign represents the maximum amount of labor hours available to produce the required level of units, which is 300 kg.

iii) The decision variables to which the constraint applies ($X_1, X_2, X_3 \dots X_n$)

iv) The impact that one unit of each decision variable will have on RHS quantity of the constraint or on the objective function (i.e., parameters). In $2X_1 + 3X_2 + 6X_3 \leq 300$ hours, of 300 hours each unit of X_1 produced utilizes 2 hours.

Characteristics of LP Models (or Assumptions in LP)

1. Well defined objective- Single objective at a time, measurable or quantifiable.
2. Finiteness of resources- resources are limited, due to which the quantity to be produced cannot be expanded infinitely.
3. Divisibility- though it is difficult to produce fraction of output, in some case it is possible.
4. Linearity of objective function and constraints- Objective function and constraints are linear.
5. Certainty- the parameters are known with certainty, due to which LP is called deterministic model. If parameters are uncertain, it will be probabilistic or stochastic model.
6. Non- negativity- the values of decision variables cannot assume negative values.

Steps in formulating LP Models

Before finding a solution for a certain LP models, the model must be developed first for a problem at hand. While developing a model, certain steps should be followed.

Step1. Identification of Decision variables- this involves recognition of a problem, i.e., identifying the product to be produced or service to be rendered to achieve the objective. Decision variables are conventionally designated by $X_1, X_2, X_3 \dots X_n$.

Step2. Determination of objective function- This involves whether the objective of the organization is maximization or minimization of certain aspect. (Maximization in this case)

Step3. Identification of constraints- those conditions which put restrictions on the level of decision variables should be identified.

Step4. Determining the parameters of constraints or objective function. This involves determination of the amount of each constraint consumed by each decision variable and/or it involves the contribution of each decision variable to objective function.

Step 5 Building and Validating the Models:- After performing step1 – step 4, the next step will be to put them in a model form and check whether any of the requirements are violated or not.

Example3.3

ABC private limited company is engaged in the production of power and traction transformers. Both of these categories of transformers pass through three basic processes: core preparation, core to coil assembly, and vapor phase drying. A power transformer yields a contribution of Birr 50,000 and traction transformer contributes Birr 10,000. The time required in the production of these two products in terms of hours for each of the processes is as follows.

Power transformer	Traction Transformer	
Core preparation	75	15
Core to Coil Assembly	160	30
Vapor Phase Drying	45	10

If the capacities available are 1000, 1500, and 750 machine hours in each processes respectively, formulate the problem as LP?

Example3.4

Lounge chairs and swivel chairs are made using the equipment of two departments as follows.

		Hours Requirements per unit		
Chair	Number of units made	Profit per unit	Department1	Department 2
Lounge	X_1	Birr 1	1	1
Swivel	X_2	1.5	1	2

Hours available in each department are 4 and 6 respectively. Develop a LP model

Solutions to LP problems

LP model can be stated in terms of two decision variables or multiple decision variables. When LP problem involve only two products, it can be solved by graphical procedure, i.e., two dimensional graphs; but if a LP problem involves a multi product case, a more advanced technique, i.e., simplex (algebraic) technique can be applied.

Graphical Method of Solving LP Problems

When linear equation or inequality involves two variables only, the solution can be described graphically. The graphical approach provides an effective visual frame of reference, and extremely helpful in understanding the kinds of phenomena which can occur in solving LP problems. From what is said above, graphical approach of solving LP problem is limited to the case of two products (decision variables). Corner or extreme point graphic method is mainly used to solve such problems and it states that for problems that have optimal solutions, a solution will occur at the corner point in the case of unique solution, while in the case of multiple solutions, at least one will occur at a corner point as these multiple solutions will be combinations of those points between two corner points.

Steps in Solving LP problems using Graphic Approach

Step 1. Form a mathematical model for the problem if the problem is in word problem form

Step 2. Graph the constraints to determine feasible region.

- ✓ If constraints are put in inequality, change the inequality sign to equality sign regardless of the original algebraic sign.
- ✓ Determine X and y intercept of each constraint
- ✓ Plot the values of intercepts on the graph
- ✓ Determine the feasible region of each constraint
- ✓ Determine the feasible region that is **common** to all the constraints.

- Feasible region- the region bounded by the operational constraints and non negativity constraints.
- Feasible solution-is a value which satisfies all the constraints (operational constraints and non negativity constraints) in a particular model.

Note: Feasibility is related with constraints, not objective function. Objective function do not tell us about the feasibility of solution.

Example 3.5

$$Z_{\max} = \text{Maximize: } 4X_1 + 7X_2 + 5X_3$$

$$2X_1 + 3X_2 + 6X_3 \leq 300 \text{ hrs ...labor hr constraint}$$

$$2X_1 + 3X_2 \leq 200 \text{ kg ...row material constraints}$$

$$X_3 \leq 30 \text{ units}$$

Assume $X_1 = 20$, $X_2 = 50$, and $X_3 = 5$

To check whether this solution is feasible or not, we use constraints, not objective function.

Hence,

For labor constraint:

$$2X_1 + 3X_2 + 6X_3 \leq 300 \text{ hrs,}$$

$$\Rightarrow 2(20) + 3(50) + 6(5) \leq 300 \text{ hrs}$$

$$\Rightarrow 220 \text{ hrs} \leq 300 \text{ hrs} \longrightarrow \text{a true statement}$$

For raw materials constraints:

$$2(20) + 3(50) \leq 200 \text{ kg}$$

$$300 \text{ kg} \leq 200 \text{ kg} \longrightarrow \text{a false statement}$$

Since the above solution does not satisfy the second constraint though it satisfies the first constraint, it is not feasible solution.

If we substitute these values to objective function:

$$Z_{\max} = 4X_1 + 7X_2 + 5X_3,$$

$$\Rightarrow 4(20) + 7(50) + 5(5)$$

$$\Rightarrow \text{Birr } 455, \text{ which does not tell us anything about feasibility of the solution.}$$

Step3. Determine the coordinates of each corner point on the region of feasible solutions.

Corner points:- points which are found at the boundary of feasible region at the intersection of two **constraints**.

Step4. Substitute the coordinates of the corner points in to the objective function to determine the corresponding value of objective function. To do so, make a table listing the value of objective function at each corner point.

Step5. Select the optimal solution. An optimal solution occurs in maximization problem at a corner point yielding the highest value of Z and in minimization problem it is found at the corner point yielding the lowest value of Z.

- ✓ Optimal point is the point (X, Y) for which the profit will be maximized or cost will be minimized.

- ✓ Optimal solution is any point in the feasible region that produces the optimal value of the objective function over the feasible region
- ✓ If the optimal value of the objective function in a LP problem exists, then the value must occur at one or more of the corner points of the feasible region.
- ✓ If the feasible region for LP problems is bounded, then both the maximum and minimum values of the objective function always exist.
- ✓ If the feasible region for LP problems is unbounded and the coefficients of the objective function are positive, then the minimum value of the objective function exists, but the maximum value does not.
- ✓ If the feasible region is empty (there are no points satisfying all the constraints), then both the maximum and minimum value of the objective function does not exist.

Graphic Method of Solving Maximization Problems

Maximization of objective function involves finding the point where the combination of products results in maximum value of objective function. The constraints are connected with $\leq$ sign. The solution space lies below the slant line and is bounded by the line segments. The origin and other points below the slant lines are in the solution space (i.e., feasible region).

Example 3.6

$$Z_{\max} = 6X_1 + 5X_2$$

Subject to:

$$2X_1 + 3X_2 \leq 300 \text{ hrs}$$

$$2X_1 + X_2 \leq 200 \text{ kg}$$

$X_1, X_2 \geq 0$, What should be the combination of decision variables (X_1 and X_2) so that the objective function will be optimum?

Example 3.7

Penile House and Furniture manufacturer produces two products: Beds and Chairs. Each unit of Bed requires 3 hrs in molding unit, 4hrs in painting unit, and 1 hr in finishing. On the other hand, each unit of Chair requires 3 hrs in molding unit, 2 hrs in the paint shop and 2 hours in finishing. Each week, there are 210 hrs available in molding, 200hrs in painting, and 120 hrs in finishing unit. The demand for Beds cannot exceed 40 units per week. Each unit of Bed contributes Birr 20 to profit, while each unit of chair contributes Birr 30. Determine the number of units of each product per week to maximize the profit? (Use Graphic Method to solve the given LP problem).

Infeasible or no solution case

Infeasibility is a condition that arises when *no value of the variable satisfy all the constraints simultaneously*. This means, there is no unique feasible region. Such a problem arises due to wrong model formulation with conflicting constraints.

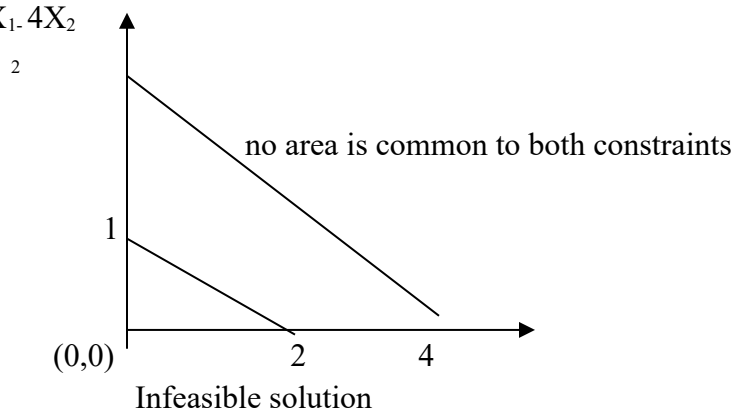
Example 3.10 Maximize $Z = 6X_1 - 4X_2$

Subject to:

$$2x_1 + 4x_2 \leq 4$$

$$4x_1 + 8x_2 \geq 16$$

$$X_1, x_2 \geq 0$$



The Simplex Method of Solving Linear Programming Models

The difference between graphic and simplex method is that the former is applicable only for problems consisting of two variables, while the later can be applicable to any number of variables. This shows that the graphic method is used for simpler problems, while simplex approach is used even for complex problems. On the other hand, simplex method consists of more variables than that of the graphic method and it can be more effective to use it when the numbers of constraints become larger and larger. Therefore, in this section we will discuss about how to solve LP problems using simplex approach for maximization problems.

The Simplex Method: Maximization with $\leq$ constraints

Simplex method or algebraic approach starts with a corner that is in the solution space, and then move to another corner until no corner can be found that will further improve the objective function. This works only with corners of the solution space and stops at the optimum corner, where no further improvement can be made in the objective function.

Basic Concepts

1. Slack variable: is the amount of resource which is unused in a given problem. Slack variables are positive constants added to operational constraints, which are linear inequalities, to convert them to linear equations.

Note:

- When resources are fully utilized, the slack will be zero and when the resources are not used at all, the slack will be the right hand side quantity (RHS), i.e. the quantity of the resource. Therefore, the value of slack lies between zero and RHS. Hence, $0 \leq \text{Slack} \leq \text{RHS}$.
- Since slack resource does not produce any product, the slack variable ordinarily contribute nothing to profit function. So, profit function of the model developed earlier will be:

$$Z_{\max} = 60x + 50y + 0s_1 + 0s_2 + 0s_3$$

2. Basic and non-basic Variables

a. Basic Variables - are variables that are in the solution with non- zero values

$$\begin{array}{lcl} 4x + 10y + s_1 = 100 & \Rightarrow & s_1 = 100 - 4x - 10y \\ 2x + y + s_2 = 22 & \Rightarrow & s_2 = 22 - 2x - y \\ 3x + 3y + s_3 = 39 & \Rightarrow & s_3 = 39 - 3x - 3y \end{array}$$

↑ basic variables at least in the initial phase and form basis in the simplex table.

b. Non-basic variables- are variables with a value of zero. They are n-m zero variables, where n is number of total variables including slacks and m is number of constraints. In our case, $n=5$, $m=3$, and non basic variables $= 5-3=2$, i.e., x and y.

3. Basic feasible solution- is a solution on the boundary of the feasible solution space, (vertex at origin). Hence, $x=0$, $y=0$. This indicates that no product is produced and the solution is called basic feasible solution, where, $x=0$, $y=0$, $s_1=100$, $s_2=22$, $s_3=39$ and profit = 0. This vertex serves as the important purpose of providing an initial basic feasible solution. It is, however merely a starting point.

4. Feasible solution- is the value which satisfy all constraints.

5. Optimal feasible- solution is a feasible solution which optimizes the objective function.

Example 3.9

A computer assembly firm is about to start production of two new micro computers which require assembly time, inspection time and storage space. The amounts of each of these resources that can be devoted to the production of the microcomputer is limited. The production manager of a firm would like to determine the

quantity of each microcomputer to be produced in order to maximize the profit generated from sales of the microcomputer.

	Type x	Type y	Maximum available resource
Profit/unit	60	50	
Assembly time	4	10	100
Inspection time	2	1	22
Storage space	3 ft ³	3 ft ³	39 ft ³

Formulate and solve the problem using simplex approach

Steps in solving LP problems using Simplex approach

Step1. Standardization of the LP model

This involves converting the inequalities in the problem statement in to equalities. To do this, we introduce new variable called slack variables to constraints and adding the same number of slack variables to the objective function, which discussed earlier. The number of slack variables equals the number of operational constraints. Hence, the standardized LP model will be:

$$Z_{\max} = 60x + 50y + 0s_1 + 0s_2 + 0s_3$$

Subject to:

$$4x + 10y + s_1 + 0s_2 + 0s_3 = 100$$

$$2x + y + 0s_1 + s_2 + 0s_3 = 22$$

$$3x + 3y + 0s_1 + 0s_2 + s_3 = 39$$

$$x, y, s_1, s_2, s_3 \geq 0$$

Step2. Develop the initial Simplex solution/ tableau.

General format

Basic Variables	C _i	Decision and slack variables	b
		Coefficient of the DV and SV	
		Parameters	RHS
	Z _i		
	C-Z		

BV		X ₁	X ₂ ... X _n	S ₁	S ₂ ... S _n	b
		C ₁	C ₂ ... C _n	0	0 ... 0	
S ₁	0	a ₁₁	a ₁₂ ... a _{1n}	1	0 ... 0	b ₁
S ₂	0	a ₂₁	a ₂₂ ... a _{2n}	0	1 ... 0	b ₂
.	.	.	.	.	.	.
S _m	0	a _{m1}	a _{m2} ... a _{mn}	0	0 ... 1	b _m
	Z					
	C-Z					

m

$$Z_i = \sum_{j=1}^n a_{ij} C_j$$

i=1

Where, i=# of rows

j=# of columns

C=coefficient of basic variable

b= constant (RHS)

The row Z values are determined column by column. For each column, the Z value is obtained by multiplying each of the members in the column by respective row coefficients in the column 'C', whereas the C-Z row which is at the bottom row that is column by column contains values that are in row Z subtracted from the values of row C.

Initial Requirements to be fulfilled to Solve LP problems

- If the order of the coefficient matrix of the operational (structural) constraints is $m \times n$, then there must be m columns of the identity matrix of order n .
- The elements of the constraint matrix of operational constraint must be non negative.
- The entries in the last row excluding the last entries, which are below the identity matrix, have to be zero.

BV	C	x	y	s ₁	s ₂	s ₃	b
S1	0	4	10	1	0	0	100
S2	0	2	1	0	1	0	22
S3	0	3	3	0	0	1	39
Z	0*	0**	0	0	0	0	
C-Z		60	50	0	0	0	

*is obtained as $(0 \times 4) + (0 \times 2) + (0 \times 3) = 0$

** is obtained by $(0 \times 10) + (0 \times 1) + (0 \times 3) = 0$

All others are calculated in the same way.

Solution at initial table:

$$S1=100$$

$$S2=22$$

$$S3=39$$

$$X=0$$

$$Y=0$$

Since x and y are not found in the BV column, the value of these variables is zero and are called non basic variables.

$$\begin{aligned} Z_{\max} &= 60x + 50y + 0s_1 + 0s_2 + 0s_3 \\ &= 60(0) + 50(0) + 0(100) + 0(22) + 0(39) \\ &= 0 \end{aligned}$$

Step 3. Test the initial tableau for optimality. The simplex optimality rule states that, to determine if the current solution is optimal, check the row C-Z values in the tableau. If any value in the C-Z value is positive, this shows that the solution is not optimal and shows the possibility for further improvement by moving to the next corner point. If the values in the C-Z are zero or negative, this is the indication of optimal solution. In our example, there are two positive values (50 and 60) in C-Z row.

Step 4. Choose the entering variable

The objective is to choose improved vertex of the feasible solution space; i.e., a vertex with a larger profit. To do this, we need to replace a basic variable with another variable. Mind you! Slack variables, which are with no contribution to objective function, are in the basic variables column, while variables with greater contribution (x and y) are out of the BV's column. Therefore, these variables should come to the BV's column to contribute to the objective function. But, which variable should come to the BV column?

Simplex incoming variable rule- states that the incoming variable is the one with the largest C-Z value. Circle the coefficients of the constraints in that columns and the column is called Pivot column. Column one is pivot column, since the largest positive number is found in the 1st column. The corresponding variable in that column is x and it is the **entering variable**.

		entering (incoming) variable ↙						
BV	C	x	y	s ₁	s ₂	s ₃	b	
S1	0	4	10	1	0	0	100	
S2	0	2	1	0	1	0	22	
S3	0	3	3	0	0	1	39	
Z		0	0	0	0	0	0	
C-Z		60	50	0	0	0	0	

↑ Pivot column

Where there is entering variable, there should be the leaving variable. But, which variable should leave the BV column so that the entering variable be part of basic variables?

Step 5. Choose the leaving variable

The simplex outgoing rule states that form ratios by dividing each entry in RHS (b) column by the corresponding entry in the pivot column of the incoming variable. Select the row with the minimum *positive ratio*, but do not form the ratio for entries in the pivot column if they have negative or zero values and the row selected is referred to as a pivot row. The element at the intersection of pivot row and pivot column is known as pivot element. This element is a key element in the succeeding computations and should always be **unity** (1). If it is different from 1, apply appropriate row operation to convert it in to 1.

		entering (incoming) variable ↙						
BV	C	x	y	s ₁	s ₂	s ₃	b	ratio
S1	0	4	10	1	0	0	100	100/4=25
S2	0	2	1	0	1	0	22	22/2=11 ← pivot row
S3	0	3	3	0	0	1	39	39/3=13
Z		0	0	0	0	0	0	
C-Z		60	50	0	0	0	0	

Leaving variable ↑ Pivot column

Leaving variable ↑ Pivot column

Note: Once the pivot row is identified, the ratios are useless and we ignore them.

Step 6. Form a new table with entering variable.

When a new variable enter in to BV column, its respective coefficient also comes in to the C column from the objective function's row(C) row. Convert pivot element to 1 by multiplying pivot row by 1/2 and convert all other elements in the same column to zero by applying the appropriate row operations.

		pivot element						
BV	C	x	y	s ₁	s ₂	s ₃	b	
S1	0	4	10	1	0	0	100	
x	60	2	1	0	1	0	22	
S3	0	3	3	0	0	1	39	
Z		0	0	0	0	0	0	
C-Z		60	50	0	0	0	0	

$-4R_2 + R_1 \rightarrow R_1$
 $1/2 R_2 \rightarrow R_2$
 $-3R_2 + R_3 \rightarrow R_3$

The resulting new table will be:

BV	C	x	y	s ₁	s ₂	s ₃	b
S1	0	0	8	1	-2	0	56
x	60	1	1/2	0	1/2	0	11
S3	0	0	3/2	0	-3/2	1	6
Z		60	30	0	30	0	660
C-Z		0	20	0	-30	0	

Solution at the second tableau

$$X=11, y=0, S_1=56, S_2=0, S_3=6$$

$$Z_{\max}=60x+50y+0s_1+0s_2+0s_3$$

$$Z_{\max}=60(11)+50(0)+0(0)+0(12)+0(9)$$

$$Z_{\max}= \text{Birr } 660$$

Note: The objective function has improved from 0 to Birr 660. Simply by looking at the solution or the BV column, it is possible to determine basic and non-basic variables. Hence, y, and S₂ are non basic variables since their value is zero and are not found in the BV column, while, x, S₁ and S₃ are basic variables since they are found in the BV column or their value in the RHS column is non-zero.

Step7. Repeat the above steps if optimal are not reached. Hence, check whether the solution in the table is optimal. Simplex approach is iterative approach. This means, the steps discussed earlier will be repeated until we reach the optimal solution.

Since there is positive value in the C-Z row (i.e., 20), this is not optimal solution. The positive value shows the possibility of further improvement of objective function.

⇒ Now identify entering and leaving variables and continue performance until you reach the optimal solution.